PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Bonus Practice C - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt both bonus problems. These are separate from the three main problems.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Textbook 15e - Question 22.52

22.52 • CALC A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

$$\rho(r) = \rho_0\left(1 - \frac{4r}{3R}\right) \qquad \text{for } r \le R$$

$$\rho(r) = 0 \qquad \text{for } r \ge R$$

where ρ_0 is a positive constant. (a) Find the total charge contained in the charge distribution. Obtain an expression for the electric field in the region (b) $r \ge R$; (c) $r \le R$. (d) Graph the electric-field magnitude E as a function of r. (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

2. [____] Textbook 15e - Question 24.36

24.36 •• Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in series to a battery that has voltage V; $C_1 = 3.00\ \mu\text{F}$ and $C_2 = 6.00\ \mu\text{F}$. The electric field between the plates of capacitor C_2 is E_{02}. While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K = 4$ is inserted between the plates of capacitor C_1, completely filling the space between them. After the dielectric is inserted in C_1, the electric field between the plates of capacitor C_2 is E_2. (a) What is the ratio E_2/E_{02}? When the dielectric is inserted into C_1, does the electric field in C_2 increase, decrease, or remain the same? (b) Repeat the calculation in part (a) for the two capacitors connected to the battery in parallel.